**MMDU MULLANA OFFICIAL SYLLABUS • BPHY-001 • UNIT 1**

# Unit 1: Wave Optics & Lasers (Complete Theory, Derivations & Quantum Systems)

**Subject:** Applied Physics (BPHY-001) • **Exam Weightage:** 26 - 30 Marks • **Study Time:** 40 mins

*Huygens' wave principle, Wavefront vs Amplitude splitting, Interference in thin films & Newton's rings derivations, 12-point Fresnel vs Fraunhofer comparison, Fraunhofer single-slit phasor analysis, Diffraction grating resolving power, Polarization laws (Brewster & Malus), Einstein A and B coefficients, Population inversion, Ruby laser, and 4-level He-Ne gas laser.*

## 1. Principles of Wave Optics & Newton's Rings (Complete Derivation)

### 1. Huygens' Principle & Superposition of Waves

**Huygens' Principle** forms the physical basis of wave optics:

1. Every point on an advancing primary wavefront serves as an independent secondary source of spherical wavelets radiating forward with wave speed $c$.

2. The envelope surface tangential to these forward wavelets at any later time $t + \Delta t$ establishes the new primary wavefront.

### 2. Division of Wavefront vs. Division of Amplitude

| Dimension | Division of Wavefront | Division of Amplitude |
|---|---|---|
| **Physical Mechanism** | Wavefront from a localized source is physically divided into two by apertures/prisms. | Wave amplitude is split by partial reflection and partial refraction at boundaries. |
| **Source Geometry** | Demands a narrow pinhole or slit source. | Can utilize broad, extended light sources. |
| **Interference Fringes** | Fringes are non-localized in space. | Fringes are localized (e.g. localized at film surface). |
| **Prototypical Systems** | **Young's Double Slit (YDSE), Fresnel Biprism, Lloyd's Mirror.** | **Newton's Rings, Thin Dielectric Films, Michelson Interferometer.** |

### 3. Newton's Rings: Derivation of Ring Diameters

When a plano-convex lens of large radius of curvature $R$ rests upon an optically flat glass plate, an air film of circular symmetry and varying thickness $t$ is formed.

✅ **FUNDAMENTAL LAW / RULE:**
**Stokes' Phase Shift Theorem:**

> When light reflects at the boundary of an optically denser medium (air-to-glass interface at the flat plate), an abrupt phase reversal of $\pi$ radians occurs, introducing an additional optical path difference of $\pm\lambda/2$.

For normal incidence ($\cos r = 1$) in an air film ($\mu = 1$), the net optical path difference is:

$$\Delta = 2t - \frac{\lambda}{2}$$

From the geometry of a sphere of radius $R$ enclosing contact radius $r_n$ and film thickness $t$:

$$r_n^2 = t(2R - t) \approx 2Rt \implies 2t = \frac{r_n^2}{R} = \frac{D_n^2}{4R}$$

where $D_n = 2r_n$ is the diameter of the $n^{\text{th}}$ fringe ring.

**(A) Condition for Dark Rings (Destructive Interference)**

$$\Delta = 2t - \frac{\lambda}{2} = (2n-1)\frac{\lambda}{2} \implies 2t = n\lambda$$

$$\frac{D_n^2}{4R} = n\lambda \implies D_n = 2\sqrt{n\lambda R} \implies D_n \propto \sqrt{n}$$

**Conclusion:** Diameters of dark rings are directly proportional to square roots of consecutive natural numbers ( $0, 1, 2, 3\ldots$).

**(B) Condition for Bright Rings (Constructive Interference)**

$$\Delta = 2t - \frac{\lambda}{2} = n\lambda \implies 2t = (2n-1)\frac{\lambda}{2}$$

$$\frac{D_n^2}{4R} = (2n-1)\frac{\lambda}{2} \implies D_n = \sqrt{2(2n-1)\lambda R} \implies D_n \propto \sqrt{2n-1}$$

**Conclusion:** Diameters of bright rings are proportional to square roots of consecutive odd numbers $(1, 3, 5\ldots)$.

**(C) Measurement of Wavelength $\lambda$ & Refractive Index $\mu$**

$$\lambda = \frac{D_{n+p}^2 - D_n^2}{4pR} \qquad \mu = \frac{(D_{n+p}^2 - D_n^2)_{\text{air}}}{(D_{n+p}^2 - D_n^2)_{\text{liquid}}}$$

*Why is the center dark in reflection?* At point of contact $O$, film thickness $t = 0$. Hence path difference $\Delta = -\lambda/2$, fulfilling the exact condition for destructive interference!

## 2. Diffraction: Fresnel vs. Fraunhofer & Single Slit Phasor Derivation

### 1. Fresnel vs. Fraunhofer Diffraction (12-Point Analysis)

| # | Parameter | Fresnel Diffraction | Fraunhofer Diffraction |
|---|---|---|---|
| 1 | **Source Distance** | Source is at **finite distance** from diffracting aperture. | Source is effectively at **infinite distance**. |
| 2 | **Screen Distance** | Screen is at **finite distance**. | Screen is effectively at **infinite distance**. |
| 3 | **Incident Wavefront** | Spherical or cylindrical wavefront. | Strictly **plane wavefront**. |
| 4 | **Diffracted Wavefront** | Spherical or cylindrical. | Strictly **plane wavefront**. |
| 5 | **Optical Lenses** | No lenses required. | Requires two convex lenses (collimator & focus). |
| 6 | **Phase Variations** | Quadratic phase terms (Fresnel integrals). | Linear phase variations across aperture. |
| 7 | **Mathematical Rigor** | Fresnel zones / Cornu spiral integration. | Fourier transform / simple phasor summation. |
| 8 | **Fringe Geometry** | Shadows of aperture; non-uniform spacing. | Spectral patterns; regular minima & maxima. |
| 9 | **Center of Pattern** | Can be bright or dark based on zone count. | Always a bright **Central Principal Maximum**. |
| 10 | **Real-Life Occurrence** | Edge of razor blade, small circular obstacle. | Spectrometers, telescope apertures, diffraction gratings. |
| 11 | **Observational Ease** | Requires critical alignment, no optical bench. | Standard spectrometer telescope configuration. |
| 12 | **Energy Concentration** | Dispersed over irregular shadow fringes. | Over **95%** of energy focused in central maximum. |

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## 2. Fraunhofer Single Slit Diffraction Derivation

A slit of width $a$ is illuminated normally by a plane wave of wavelength $\lambda$. Dividing the slit into $N$ equal elements of width $dx = a/N$, the resultant amplitude $R$ at angle $\theta$ obtained by phasor addition is:

$$R = A_0 \frac{\sin \alpha}{\alpha} \quad \text{where } \alpha = \frac{\pi a \sin \theta}{\lambda}$$

The intensity distribution is:

$$I(\theta) = I_0 \left(\frac{\sin \alpha}{\alpha}\right)^2$$

- **Central Maximum ($\theta = 0$):** As $\alpha \to 0$, $\lim_{\alpha \to 0} \frac{\sin \alpha}{\alpha} = 1 \implies I = I_0$.
- **Minima Condition:** $\sin \alpha = 0$ while $\alpha \neq 0 \implies \alpha = \pm m\pi$ ($m = 1, 2, 3...$):

$$a \sin \theta = \pm m\lambda$$

- **Secondary Maxima Condition:** $\frac{dI}{d\alpha} = 0 \implies \tan \alpha = \alpha$. Solved graphically: $\alpha \approx \pm 1.43\pi, \pm 2.46\pi, \pm 3.47\pi$.
- **Intensity Ratio:** $I_0 : I_1 : I_2 : I_3 = 1 : 0.045 : 0.016 : 0.008$.
- **Central Angular Width:** $2\theta = \frac{2\lambda}{a}$ (Linear width on screen at focal length $f$ is $2f\lambda/a$).

## 3. Polarization of Light & Birefringence

### 1. Polarization Fundamentals

In electromagnetic waves, the electric field $\vec{E}$ vibrates transversely to the propagation axis.

- **Unpolarized Light:** Symmetrical vibrations in all planes perpendicular to the propagation ray.
- **Linearly (Plane) Polarized Light:** Electric field vibrations restricted strictly to a single plane.

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### 2. Brewster's Law & Polarizing Angle

When unpolarized light is incident on a transparent dielectric medium at the polarizing angle $i_p$, the reflected ray is 100% plane-polarized perpendicular to the plane of incidence.

- Reflected ray and refracted ray are mutually perpendicular: $i_p + r = 90^\circ \implies r = 90^\circ - i_p$.
- From Snell's Law: $\mu = \frac{\sin i_p}{\sin r} = \frac{\sin i_p}{\sin(90^\circ - i_p)} = \frac{\sin i_p}{\cos i_p} = \tan i_p$.

$$\mu = \tan(i_p)$$

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### 3. Malus's Law

When plane-polarized light of intensity $I_0$ passes through an analyzer oriented at angle $\theta$ relative to the transmission axis of the polarizer:

$$I = I_0 \cos^2 \theta$$

- When $\theta = 0°$ (parallel axes): $I = I_0$ (maximum transmission).
- When $\theta = 90°$ (crossed nicols): $I = 0$ (complete extinction).

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### 4. Birefringence & Retardation Plates

In anisotropic crystals (e.g. Calcite $CaCO_3$, Quartz), an incident ray splits into an **Ordinary Ray (O-ray)** obeying Snell's Law and an **Extraordinary Ray (E-ray)** with direction-dependent velocity.

- **Quarter-Wave Plate (QWP):** Introduces path difference $\lambda/4$ (phase shift $\pi/2$):

$$t = \frac{\lambda}{4(\mu_o - \mu_e)}$$

- **Half-Wave Plate (HWP):** Introduces path difference $\lambda/2$ (phase shift $\pi$):

$$t = \frac{\lambda}{2(\mu_o - \mu_e)}$$

## 4. Quantum Laser Physics: Einstein Coefficients & He-Ne Systems

### 1. Fundamental Characteristics of Laser Light

LASER stands for **Light Amplification by Stimulated Emission of Radiation**.

1. **Extreme Monochromaticity:** Spectral linewidth $\Delta\lambda \approx 10^{-6}$ Å.
2. **Directionality & Low Divergence:** Beam divergence angle $\theta < 1$ mrad.
3. **High Coherence:** Exceptional temporal coherence length (kilometers) and spatial wavefront coherence.

4. **Enormous Intensity:** Focused laser beams exceed solar core flux density ($> 10^{12}$ W/cm$^2$).

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## 2. Einstein's A and B Coefficients Derivation

Consider two atomic levels $E_1$ and $E_2$ in thermal equilibrium with radiation density $u(\nu)$ at temperature $T$:

1. **Absorption:** $R_{12} = B_{12} N_1 u(\nu)$
2. **Spontaneous Emission:** $R_{21\,\text{sp}} = A_{21} N_2$
3. **Stimulated Emission:** $R_{21\,\text{st}} = B_{21} N_2 u(\nu)$

At thermodynamic equilibrium: Rate of Upward Transitions = Rate of Downward Transitions:

$$B_{12} N_1 u(\nu) = A_{21} N_2 + B_{21} N_2 u(\nu) \implies u(\nu) = \frac{A_{21}}{B_{12}\frac{N_1}{N_2} - B_{21}}$$

From Maxwell-Boltzmann statistics: $N_1/N_2 = e^{(E_2 - E_1)/k_B T} = e^{h\nu/k_B T}$:

$$u(\nu) = \frac{A_{21}}{B_{21}} \left[\frac{1}{\frac{B_{12}}{B_{21}} e^{h\nu/k_B T} - 1}\right]$$

Comparing directly with **Planck's Law of Blackbody Radiation**:

$$u(\nu) = \frac{8\pi h \nu^3}{c^3} \left[\frac{1}{e^{h\nu/k_B T} - 1}\right]$$

We arrive at two fundamental Einstein identities:

$$B_{12} = B_{21} \qquad \frac{A_{21}}{B_{21}} = \frac{8\pi h \nu^3}{c^3} \propto \nu^3$$

*Physical Implication:* The probability of spontaneous emission increases dramatically with frequency $\nu^3$, explaining why gamma-ray and X-ray lasers require astronomical pumping powers compared to microwave MASERs.

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## 3. Population Inversion & Metastable States

Under thermal equilibrium, $N_1 > N_2$. For net optical gain, one must enforce **Population Inversion ($N_2 > N_1$)**. This is achieved by pumping atoms into a **Metastable State** having an unusually long lifetime ($\sim 10^{-3}$ s compared to normal $10^{-8}$ s).

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### 4. Four-Level Helium-Neon (He-Ne) Gas Laser

- **Active Medium:** Gas mixture of Helium and Neon in a 10:1 ratio at low pressure ($\sim 1$ torr).
- **Pumping Mechanism:** High-voltage DC electrical discharge. Fast electrons excite He atoms from ground state $1^1S_0$ to metastable levels $2^1S_0$ (20.61 eV) and $2^3S_1$ (19.82 eV).
- **Resonant Energy Transfer:** Excited He atoms collide inelastically with ground state Neon atoms, transferring energy with minimal mismatch ($\Delta E \approx 0.05$ eV) to excite Neon into $3s$ (20.66 eV) and $2s$ (19.78 eV) states.
- **Laser Emission:** Stimulated radiative transition occurs from Neon $3s \rightarrow 2p$, emitting a continuous, highly coherent red laser beam at:

$$\lambda = 632.8 \text{ nm}$$

- **Depopulation:** Fast spontaneous decay occurs from $2p \rightarrow 1s$, followed by non-radiative de-excitation to the ground state via physical collisions with the capillary tube walls.